

Computer Networks Lab

Fall 2025

Week 11

Dynamic Routing (OSPF and EIGRP)

Router in use:

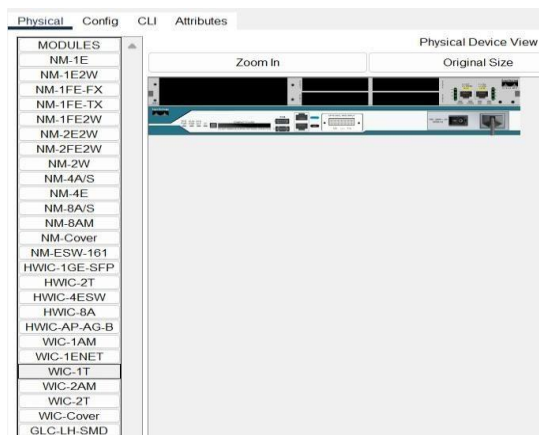


How to add and use Serial Cable:

1. Go to the physical portion of the router.
2. Turn off the router.



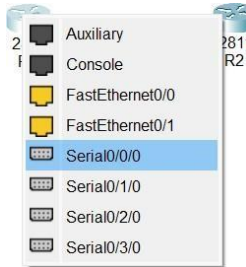
3. Choose WIC-1T



4. Add ports according to need, and turn on the router



5. To connect two routers using serial cable, use Serial DTE wire 
6. Connect one's serial port with the other router's



7. And you will have a connection like:



8. Add ip address, turn on the ports and you are good to go.

OSPF routing

Open Shortest Path First (OSPF) is a link-state routing protocol that is used to find the best path between the source and the destination router using its own Shortest Path First). OSPF is developed by Internet Engineering Task Force (IETF) as one of the Interior Gateway Protocol (IGP), i.e, the protocol which aims at moving the packet within a large autonomous system or routing domain. It is a network layer protocol which works on protocol number 89 and uses AD value 110. OSPF uses multicast address 224.0.0.5 for normal communication and 224.0.0.6 for update to designated router(DR)/Backup Designated Router (BDR).

Features of OSPF:

1. Link-State Protocol: OSPF is a link-state routing protocol. It builds a detailed topology of the entire network by exchanging link-state advertisements (LSAs) between routers.
2. Hierarchical Design: OSPF supports hierarchical network design through the use of areas. This allows for scalability and reduced routing overhead.
3. Fast Convergence: OSPF converges quickly in response to network changes due to its efficient SPF (Shortest Path First) algorithm.
4. Variable-Length Subnet Masking (VLSM): OSPF supports VLSM, allowing for the creation of subnets with different subnet masks within the same network.

5. Authentication: OSPF supports authentication mechanisms to secure routing information exchange between OSPF routers.
6. Cost-Based Metric: OSPF calculates the cost of each route based on the bandwidth of the link. This metric helps in choosing the shortest path to a destination.
7. Multiple Routing Tables: OSPF supports multiple routing tables, including one for IP routes and another for MPLS (Multi-Protocol Label Switching) traffic.
8. Dynamic Routing Updates: OSPF routers exchange routing information only when there are changes in the network topology, reducing unnecessary bandwidth consumption compared to RIP.
9. Neighbour Adjacencies: OSPF routers establish neighbour adjacencies only with directly connected OSPF routers, enhancing security and stability.
10. Route Summarization: OSPF allows for route summarization, which can reduce the size of routing tables and simplify routing in larger networks.
11. Support for IPv6: OSPF supports IPv6, making it suitable for modern networks that utilize IPv6 addressing.
12. Load Balancing: OSPF can perform load balancing by distributing traffic across multiple equalcost paths to a destination.

What is a Wildcard Mask?

A **wildcard mask** tells the router which bits of the IP address to **match or ignore** when selecting networks to advertise in EIGRP.

Comparison: Subnet Mask vs Wildcard Mask

Subnet Mask	Wildcard Mask	Meaning
255.255.255.0	0.0.0.255	Match the first 3 octets exactly
255.255.255.252	0.0.0.3	Match the first 30 bits

- **Subnet mask:** defines the range of usable IPs
- **Wildcard mask:** used to **select specific IP ranges** when configuring routing or ACLs

What is a Wildcard Mask in OSPF?

- Just like in EIGRP, wildcard masks in OSPF define **which IP networks** OSPF should operate on. - **Example:**
- network 192.168.1.0 0.0.0.255 area 0

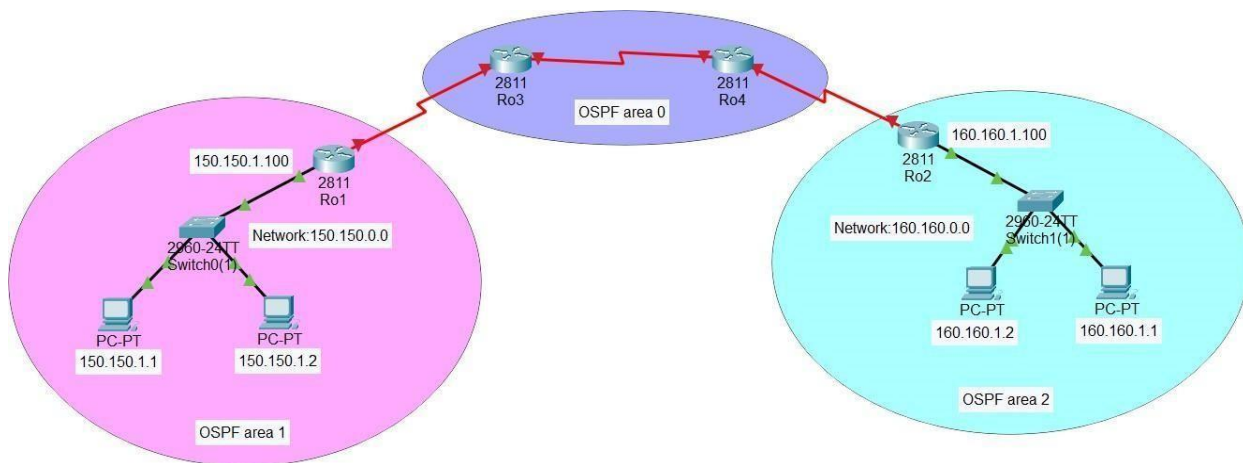
- This means:

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- Advertise the subnet **192.168.1.0/24**
- Place it in **Area 0**

Walk Through Task:

Step 1: Create a topology as such:



Step 2: Assign ip addresses to the PC accordingly as mentioned in the image above with the accurate default gateway.

Step 3: Once all the systems are assigned with an ip address, set the default gateway.

For router in network: 150.150.0.0 (pink area):

Step 3.1: go to the router's CLI

Step 3.2: type '**en**'

Step 3.3: type '**config t**' to access configuration terminal

Step 3.4: access the interface connected to the switch in the network (in this case its port 0/0)

Step 3.5: add ip address by using the command '**ip address 150.150.1.100 255.255.0.0**'.

Here 150.150.1.100 is the ip address, while 255.255.0.0 is the subnet mask.

Step 3.6: '**no shutdown**' the interface.

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 150.150.1.100 255.255.0.0
Router(config-if)#ip address 150.150.1.100 255.255.0.0
Router(config-if)#no shutdown
Router(config-if)#
```

For router in network: 160.160.0.0 (blue area):

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Step 3.1: go to the router's CLI

Step 3.2: type 'en'

Step 3.3: type 'config t' to access configuration terminal

Step 3.4: access the interface connected to the switch in the network (in this case its port 0/0)

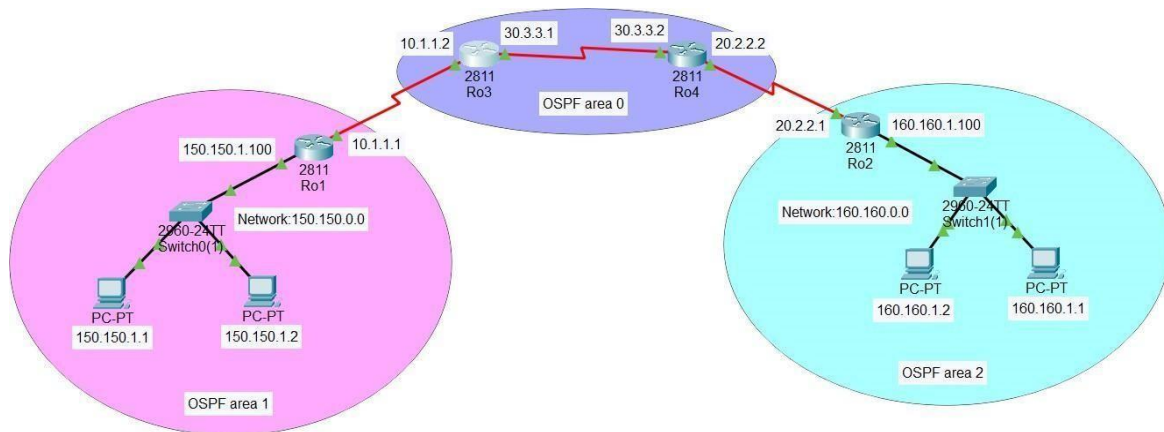
Step 3.5: add ip address by using the command 'ip address 160.160.1.100 255.255.0.0'.

Here 160.160.1.100 is the ip address, while 255.255.0.0 is the subnet mask.

Step 3.6: 'no shutdown' the interface.

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 160.160.1.100 255.255.0.0
Router(config-if)#ip address 160.160.1.100 255.255.0.0
Router(config-if)#no shutdown
Router(config-if)#
```

Repeat step 3 for the other area as well



Step 4: Set up the OSPF routing

router OSPF <process_id>

network <network_id> <wildcard mask> area <area_number>

Process id can be any number from 1 to 65525. Let us choose 10 for the process id

For the router in area 1

Step 4.1: For router in area 1

Run the following commands

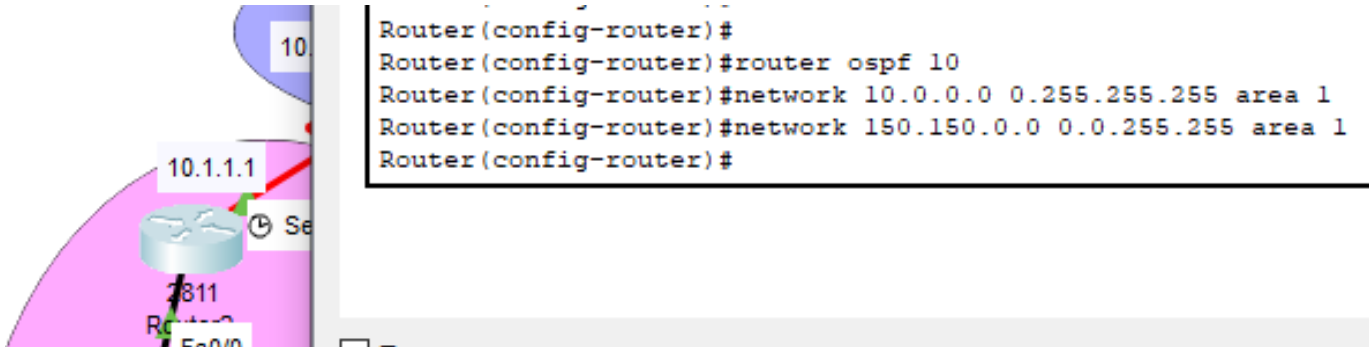
router(config)# **router ospf 10**

router(router-config)# **network 10.0.0.0 0.255.255.255 area 1**

router(router-config)# **network 150.150.0.0 0.255.255 area 1**

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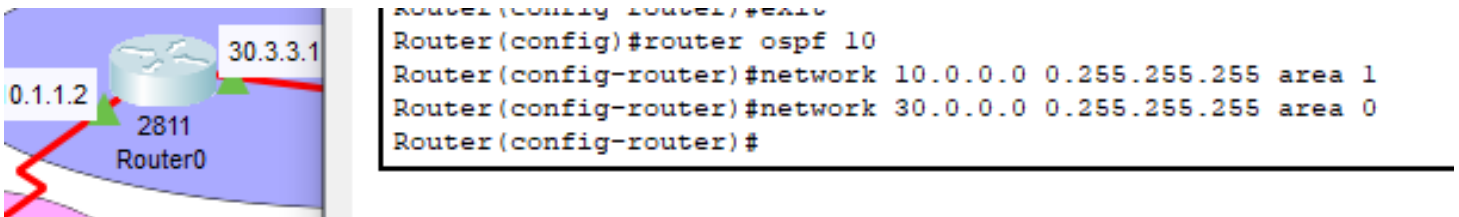
(trick here to remember is the wire going out of the network like here the wire in network 10 will not be in the backbone area)



For the router in area 0 connecting to area 1

Step 4.1: For router in area 0 Run the following commands

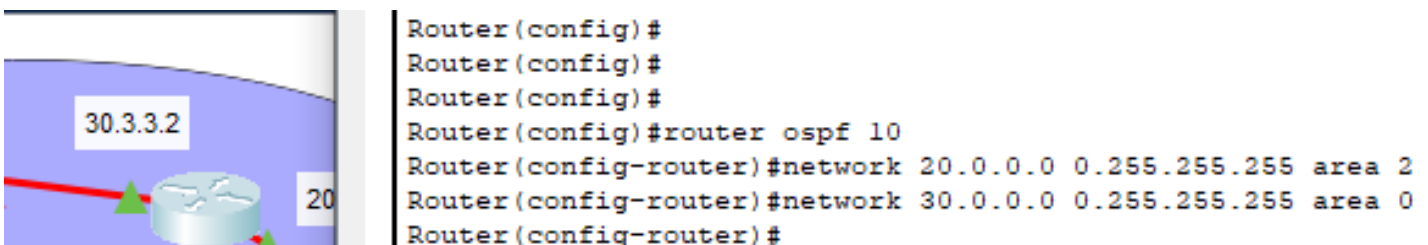
```
router(config)# router ospf 10
router(router-config)# network 10.0.0.0 0.255.255.255 area 1
router(router-config)# network 30.0.0.0 0.255.255.255 area 0
```



For the router in area 0 connecting to area 2

Step 4.1: For router in area 0 Run the following commands

```
router(config)# router ospf 10
router(router-config)# network 20.0.0.0 0.255.255.255 area 2
router(router-config)# network 30.0.0.0 0.255.255.255 area 0
```



For the router in area 2

Step 4.1: For router in area 2

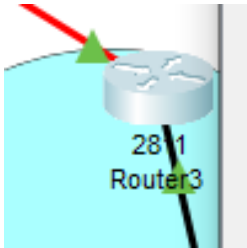
Run the following commands

```
router(config)# router ospf 10
```

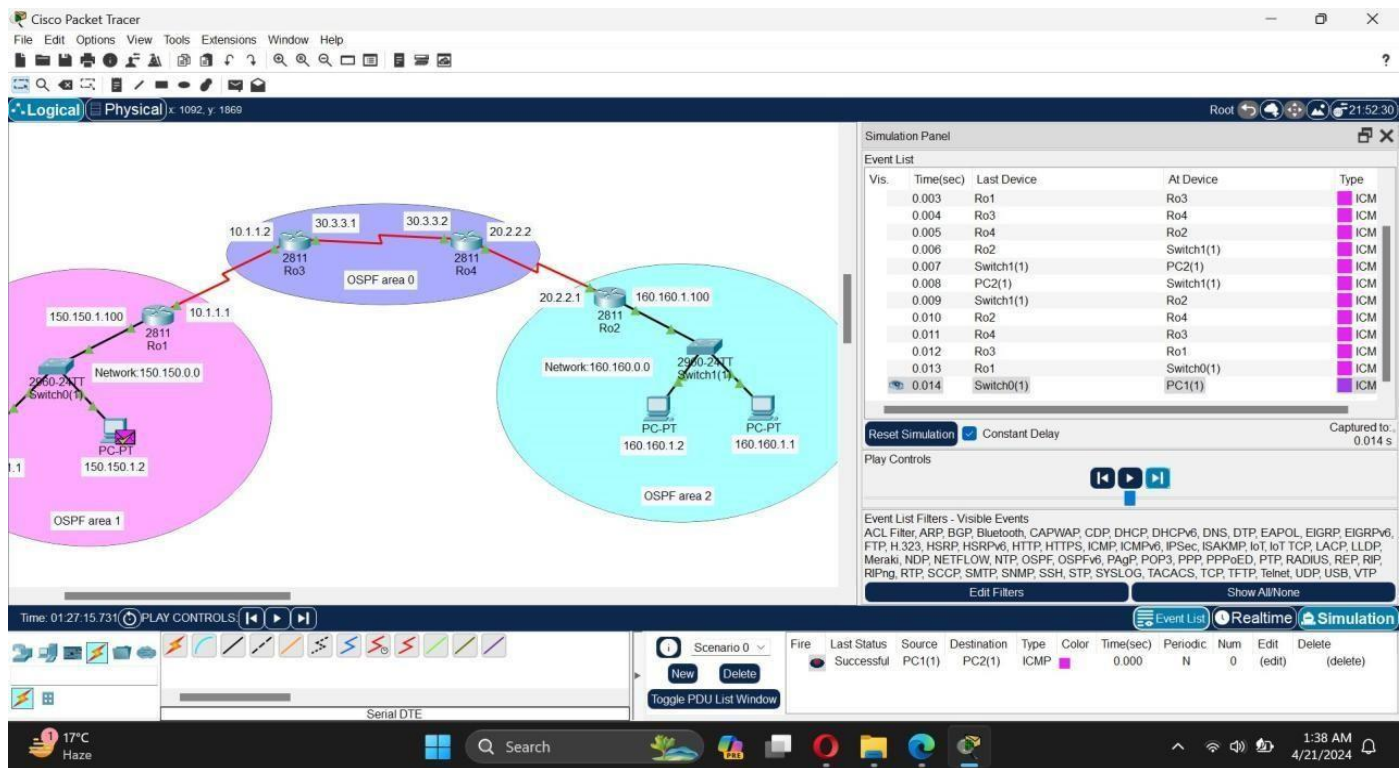


```
router(router-config)# network 160.160.0.0 0.255.255 area 2
```

```
router(router-config)# network 20.0.0.0 0.255.255 area 2
```



```
Router(config)#
Router(config)#
Router(config)#router ospf 10
Router(config-router)#network 160.160.0.0 0.0.255.255 area 2
Router(config-router)#network 20.0.0.0 0.255.255.255 area 2
Router(config-router)#
```



EIGRP Routing

Enhanced Interior Gateway Routing Protocol (EIGRP) is an advanced distance-vector routing protocol developed by Cisco. It is used to determine the best path for data transmission within a routing domain or autonomous system. EIGRP combines the best features of distance-vector and link-state protocols, making it a hybrid protocol. It operates at the network layer and uses protocol number 88, with an administrative distance (AD) of 90 for internal routes and 170 for external routes. EIGRP uses the multicast address 224.0.0.10 to communicate with neighboring routers.

Features of EIGRP:

1. Hybrid Protocol: EIGRP blends the advantages of both distance-vector and link-state protocols. It uses DUAL (Diffusing Update Algorithm) to calculate loop-free and efficient paths.

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2. Fast Convergence: EIGRP provides rapid convergence in case of network changes, thanks to DUAL which ensures backup routes are ready for immediate use.
3. Supports VLSM and CIDR: Like OSPF, EIGRP supports Variable-Length Subnet Masking and
4. Classless Inter-Domain Routing, enabling efficient IP address utilization.
5. Efficient Bandwidth Usage: EIGRP sends incremental updates instead of entire routing tables, reducing bandwidth usage.
6. Multiple Protocol Support: EIGRP supports routing for multiple network layer protocols, including IPv4, IPv6, IPX, and AppleTalk (legacy).
7. Composite Metric: EIGRP uses a metric based on bandwidth, delay, reliability, and load, offering more flexible and accurate route selection.
8. Neighbor Discovery and Maintenance: EIGRP maintains relationships with directly connected routers using Hello packets, enabling quick detection of topology changes.
9. Loop-Free Topology: Using the DUAL FSM, EIGRP ensures loop-free routes without relying on traditional methods like hold-downs or split horizon.
10. Route Summarization: EIGRP supports both manual and automatic route summarization, which helps reduce routing table size and optimize performance.
11. Unequal-Cost Load Balancing: EIGRP can perform load balancing across unequal-cost paths using the variance command — a unique feature compared to most IGPs.
12. Authentication: EIGRP supports authentication of routing updates to enhance the security of routing information exchanges.
13. IPv6 Support: EIGRP supports IPv6, making it future-ready and compatible with modern IP addressing schemes.

What is an Autonomous System Number (ASN)?

An Autonomous System (AS) is a group of routers under a single administrative control that shares routing information.

- Think of it like an organization's internal network.
- When configuring EIGRP, you specify an ASN to identify routers that belong to the same EIGRP group.
- Routers with the same ASN form EIGRP neighbor relationships.

Example:

- `router eigrp 100` This means you're starting EIGRP with Autonomous System Number 100.
- If Router A and Router B both use ASN 100, they'll talk EIGRP and share routing info.
- If one uses 100 and another uses 200, they won't become EIGRP neighbors.

Wildcard Mask in EIGRP:

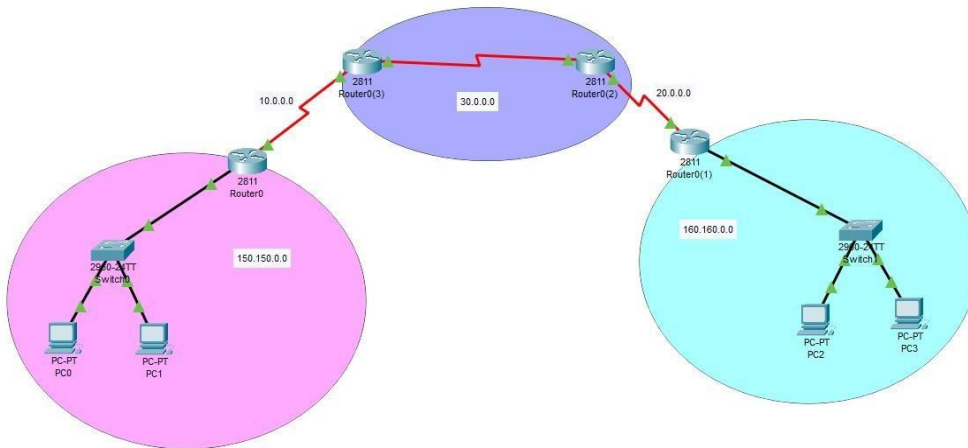
In EIGRP, you can specify a wildcard mask to be **precise about which networks** the router should include in EIGRP updates.

`router eigrp 1 network 192.168.1.0 0.0.0.255` This means:

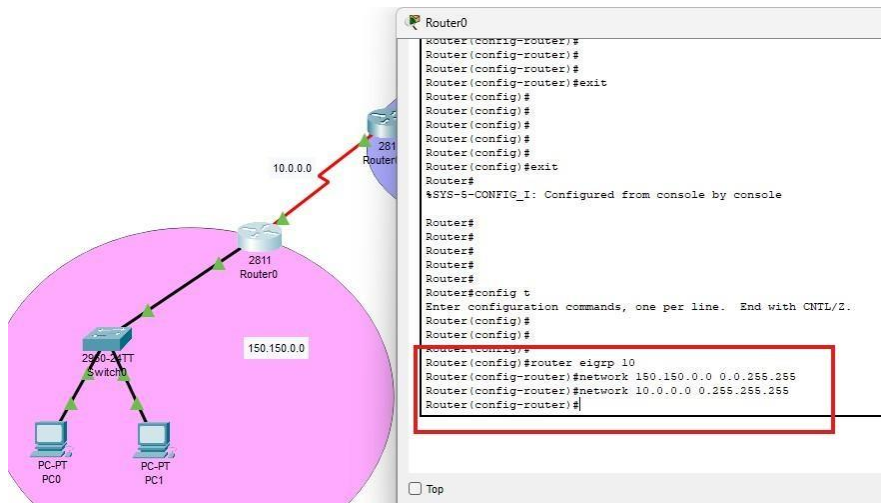
- Advertise all IPs **from 192.168.1.0 to 192.168.1.255**
- Because **0.0.0.255** says: *match the first 3 octets, ignore the last*

Walk through :

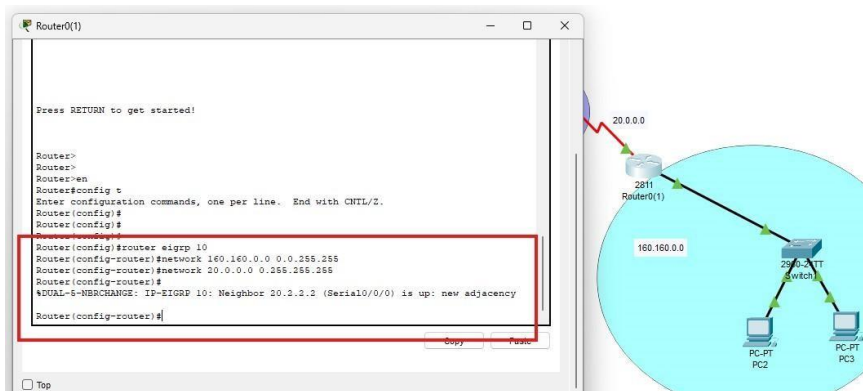
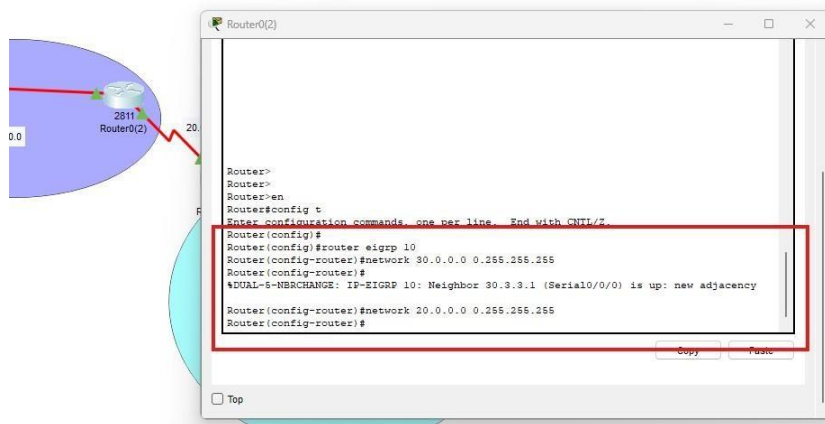
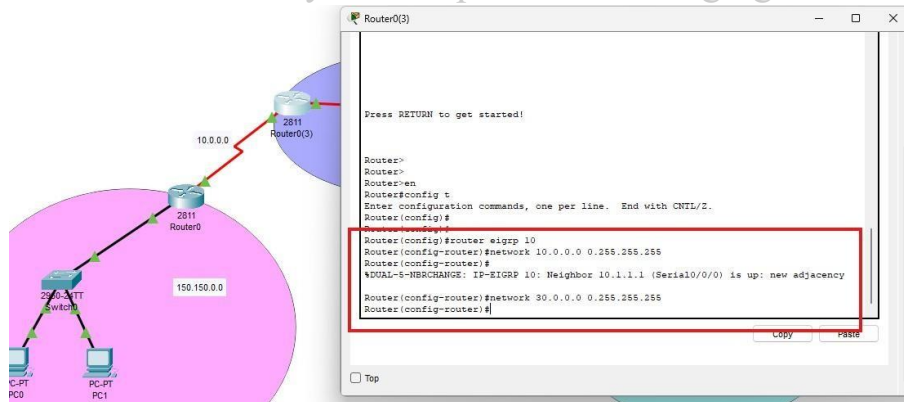
We will be using the same topology as we used for OSPF.



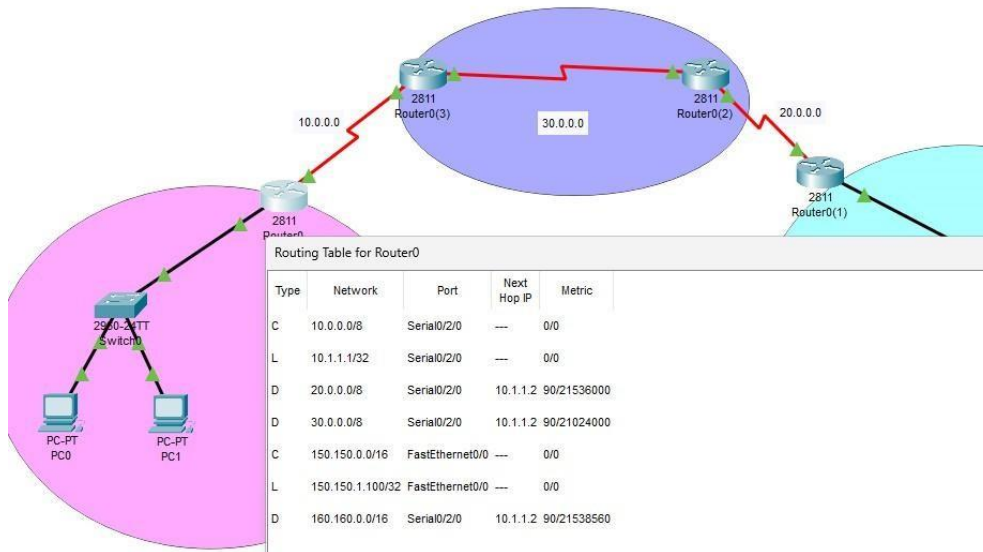
Step 1: Go to the first router and run the following commands for EIGRP:



Step 2: Repeat the same for the other routers as well



Step 3: Check the routing tables for all the routers to confirm the configuration.



Lab tasks

Consider your roll number

Let xx be the last two digits of your roll number

Example: for roll number 22i0011

11 be xx and 12 will be xx+1

Task 1:

Construct a network topology and implement OSPF and EIGRP separately.

Ip addressing:

Network A: 162.xx.0.0

Network B: 162.xx+1.0.0

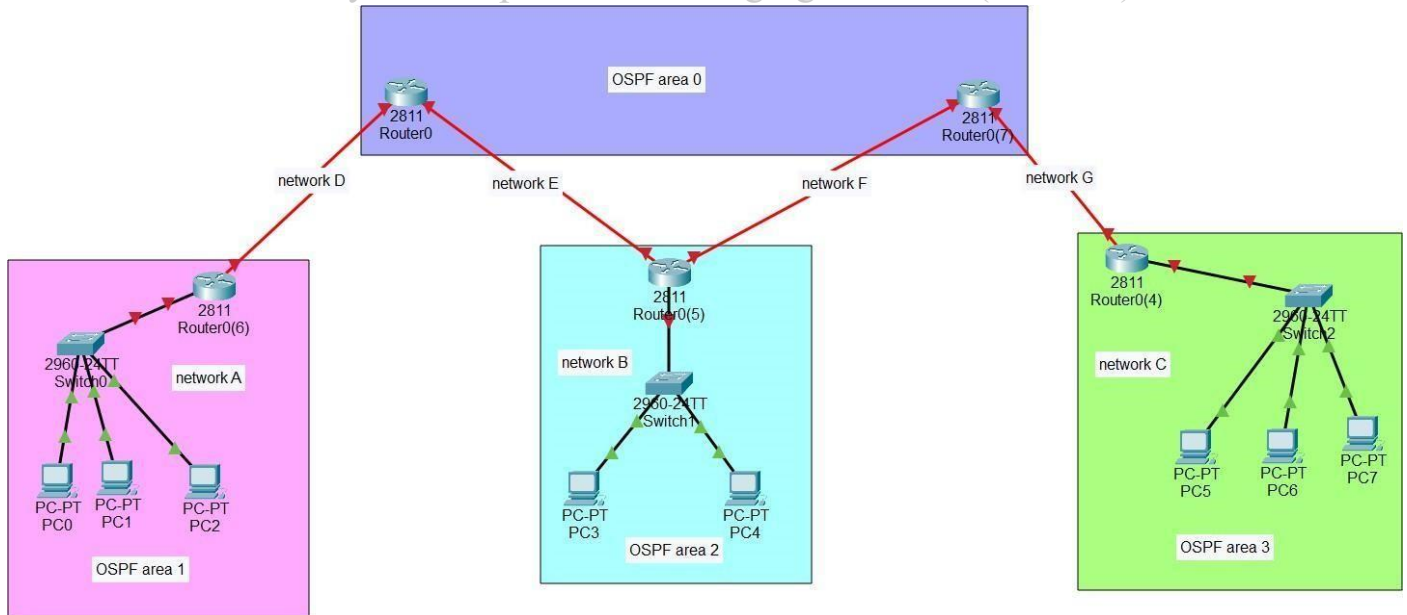
Network C: 162.xx+2.0.0

Network D: 162.xx+3.0.0

Network E: 162.xx+4.0.0

Network F: 162.xx+5.0.0

Network G: 162.xx+6.0.0



Submission Guidelines:

- Submit the .pkt file , renamed to your roll number with task number.
- Submit a document containing screenshots of topology diagrams, routing table screenshots, and successful message transfer between different networks.